BED-BPP DataSet

https://floriankagerer.github.io/dataset/

We present our benchmarking dataset for robotic bin packing, called BED-BPP. We aim to provide a dataset that is used to compare the performance of algorithms that solve many variants of the three-dimensional bin packing problem.

Whenever you use the dataset, please cite our publication:

> *Kagerer F, Beinhofer M, Stricker S, Nüchter A. BED-BPP: Benchmarking dataset for robotic bin packing problems. The International Journal of Robotics Research. 2023;42(11):1007-1014. doi:10.1177/02783649231193048*
> *[BibTeX] [DOI]*

Description

This section briefly describes the dataset BED-BPP. For a detailed description of the dataset, please read our publication. The usage of the dataset is shown in our Git Repository bed-bpp-env.

Characteristics. The following table summarizes the characteristics of the dataset.

Property	Value (Mean ± Standard Deviation)
length	10 003 orders
different articles	2 621 articles
different item sizes	974 sizes
items per order	43 items ± 14 items
length of longest n-gram	8 items ± 3 items
different item sizes per order	22 sizes ± 8 sizes

Structure. In general, the dataset BED-BPP has the following structure:

```
{
    "order_i": {
        "item_sequence": {
            "1": {
                "article": "article-id1",
                "id": "id1",
                "product_group": "product_group1",
                "length/mm": 300,
                "width/mm": 200,
                "height/mm": 100,
                "weight/kg": 3.14,
                "sequence": 1
            },
            "2": {...},
            ...
        },
        "properties": {
            "id": "0123456",
            "order_nr": "01234567",
            "type": "chilled frozen grocery",
            "target": "euro-pallet"
        }
    }
}
```